

CSE 4616 – Wireless Networks

Lab 02: In-Lab Assessment Tasks

Duration: 1 hour 30 minutes — Individual Assessment

Tools: OMNeT++ and INET Framework

General Instructions

- All tasks start from the **ChannelBaseline** configuration defined in the lab manual, using the same NED file as Lab 1.
- Tasks 1–3 require **INI file changes only**. Task 4 requires both a NED and an INI file change.
- Name your NED files: `TaskX_StudentID_Lab2.ned`
- Take Qtenv screenshots while the simulation is running unless stated otherwise.
- Show all calculations in your lab report. Reference Lecture 1 concepts in your written answers.

Baseline Configuration Reminder:

- Network: `WiredAndWirelessHostsWithAP` (Lab 1 NED file, unchanged for Tasks 1–3)
- 3 hosts + 1 router + 1 AP; `wirelessHost1` is the only wireless device
- Packet size: 512B, send interval: 0.5s, sim limit: 200s
- `txPower = 20mW`, `bitrate = 54Mbps`, `sensitivity = -85dBm`
- Path loss: `FreeSpacePathLoss`, `noise = -110dBm`

Task 01: Signal Attenuation and Transmission Range

Background: In Lecture 1 you learned that the maximum usable range of a wireless link is determined by the transmit power, the receiver sensitivity, and the path loss model. When the path loss between transmitter and receiver exceeds the maximum tolerable value, the received signal falls below the sensitivity threshold and packets are lost. In this task you will vary the transmit power and observe how the effective communication range changes.

Required Modifications (INI file only):

Create **three** configuration sections, each with a different transmit power:

1. `Power_High` with a transmission power of 100mW
2. `Power_Medium` with a transmission power of 5mW
3. `Power_Low` with a transmission power of 0.5mW

Run each configuration and record your observations.

Calculation Exercise:

For each transmit power level, calculate the **maximum tolerable path loss** and the corresponding **maximum range** in free space at 2.4GHz. Use receiver sensitivity = -85 dBm and the formula from the lab manual. Show all working.

Config	Power (mW)	Power (dBm)	Max Path Loss (dB)	Max Range (m)
Power_High	100			
Power_Medium	5			
Power_Low	0.5			

Table 1: Range calculation table for Task 01 — complete all cells

Analysis Questions:

- a. In Qtenv, did `Power_Low` result in packet loss or complete link failure? Describe exactly what you observed in the simulation (packet animations, console messages).
- b. From Lecture 1: “*The received signal must have sufficient strength above the noise to be received without error.*” Using your calculated maximum ranges, explain why reducing transmit power by a factor of 10 does *not* reduce the range by a factor of 10.
- c. A device manufacturer wants to double the wireless range without changing the receiver sensitivity. By how many dB must the transmit power increase? Show the calculation.
- d. In `Power_Low`, did the wired link between `wiredHost1` and `wiredHost2` continue to work normally? Why or why not? Connect your answer to the Lecture 1 comparison of wired vs. wireless attenuation.

Deliverables

- **INI file:** `omnetpp.ini` (NED file unchanged from Lab 1)
- **Three screenshots:** one per power configuration, taken at $t \approx 60$ s
- **Completed calculation table**
- **Written answers** to (a)–(d)

Task 02: SNR Degradation and the Noise Floor

Background: From Lecture 1: “Lower signal-to-noise ratio (SNR) increases the probability of bit errors.” The noise floor in a wireless system is set by thermal noise, interference from other devices, and environmental electromagnetic radiation. In this task you will raise the background noise level progressively and observe how the SNR at the receiver degrades, leading to increased packet errors.

Required Modifications (INI file only):

Create **four** configurations with increasing noise levels:

1. Noise_VeryLow with a background noise of -120dBm
2. Noise_Baseline with a background noise of -110dBm
3. Noise_High with a background noise of -90dBm
4. Noise_Extreme with a background noise of -75dBm

SNR Calculation Exercise:

Assume wirelessHost1 is approximately 10 m from the access point. At 2.4 GHz in free space, the path loss over 10 m is approximately 60 dB.

Given `transmitter.power = 20mW = 13 dBm`:

$$P_{received} = 13 \text{ dBm} - 60 \text{ dB} = -47 \text{ dBm}$$

Complete the SNR margin table below:

Configuration	Noise (dBm)	Floor	SNR at 10m (dB)	Expected behaviour
Noise_VeryLow	-120			
Noise_Baseline	-110			
Noise_High	-90			
Noise_Extreme	-75			

Table 2: SNR calculation table for Task 02

For the **Expected behaviour** column, refer to the modulation SNR table in the lab manual. State which modulation would still work at each SNR level (e.g., “54 Mbps viable”, “only 6 Mbps viable”, “link fails”).

Analysis Questions:

- a. In `Noise_Extreme`, what happened to traffic from `wirelessHost1`? Did wired traffic between `wiredHost1` and `wiredHost2` also fail? Explain why or why not.
- b. From Lecture 1: “*Noise (thermal, impulse) distorts the signal, making it hard to detect correct bits.*” Using your SNR table, at which noise level does the 54 Mbps link first become unreliable? Justify using the lab manual’s SNR–modulation table.
- c. In `Qtenv`, run `Noise_High` and `Noise_Extreme` back to back. Describe any visual difference in packet animations between the two runs.
- d. A real-world example from Lecture 1 is a 2.4 GHz microwave oven interfering with Wi-Fi. Which of your four configurations most closely models this scenario? Explain your reasoning.

Deliverables

- **INI file:** `omnetpp.ini`
- **Four screenshots:** one per configuration at $t \approx 60$ s
- **Completed SNR table**
- **Written answers** to (a)–(d)

Task 03: Path Loss Model Comparison

Background: In Lecture 1 you learned that in free space signal power falls as $1/d^2$, but in real environments the decay can be more severe due to reflections, diffraction, and scattering. The **LogNormal Shadowing** model captures the random variations in signal strength caused by obstacles and environmental clutter, while the path loss exponent α controls the average rate of decay. In this task you will observe how different values of α affect the effective range and packet delivery in realistic environments.

Required Modifications (INI file only):

Create the following configurations:

1. `PathLoss_FreeSpace` using the `FreeSpacePathLoss` model
2. `Multipath_Study` using the `LogNormalShadowing` model with path loss exponent $\alpha = 3.5$
3. `PathLoss_EdgeOfCoverage` using the `LogNormalShadowing` model with:
 - path loss exponent $\alpha = 3.5$
 - transmitter power reduced to 5mW
 - receiver sensitivity set to -85dBm

For each configuration, modify only the parameters related to:

- path loss model selection
- path loss exponent
- transmitter power
- receiver sensitivity

Observe how increasing path loss and reducing transmission power affect communication range and packet delivery performance.

Analysis Questions:

- a. From Lecture 1: *“Receiving power is proportional to $1/d^2$ in vacuum but much more in real environments.”* The LogNormal Shadowing model introduces random variations around the mean path loss. If a device at 20 m receives -60 dBm under free space, estimate the expected received power at 40 m using the path loss exponent $\alpha = 3.5$ (ignore random shadowing for the calculation).
- b. In `Qtenv` or `Cmdenv`, compare the `PathLoss_FreeSpace` and `PathLoss_LogNormalShadowing` runs. Did you observe any difference in packet delivery from `wirelessHost1`? If not, explain why the node distances may be too small to reveal significant shadowing effects.
- c. In `PathLoss_EdgeOfCoverage`, describe what happened to traffic from `wirelessHost1`. Was it completely lost or intermittent? How does shadowing contribute to variability in reception?

- d. From Lecture 1, reflection, diffraction, and scattering all contribute to multipath. Which of these mechanisms does the LogNormal Shadowing model implicitly capture, and which does it ignore? What type of environment is the LogNormal Shadowing model most appropriate for?

Deliverables

- **INI file:** `omnetpp.ini`
- **Three screenshots:** one per configuration at $t \approx 60$ s
- **Written answers** to (a)–(d)

Task 04: Co-Channel Interference — NED + INI Extension

Background: In Lecture 1 you studied that wireless is a shared broadcast medium and that “*interference from other sources*” is one of the key differences from wired networks. A real example given in the lecture is a 2.4 GHz wireless phone interfering with an 802.11b network. In this task you will model this by adding an **interferer** host that transmits on the same channel as `wirelessHost1` and observing the resulting SINR degradation.

Required Modifications:

NED file: Add an additional wireless host submodule named `interferer` inside the `submodules` block of the `WiredAndWirelessHostsWithAP` network. Position the module near the existing wireless nodes using an appropriate display annotation.

INI file:

Create the following two configurations:

1. `Interference_Control`

- Configure the network to run without active interference
- Keep the `interferer` node present in the topology
- Disable all applications on the `interferer` node so that it remains silent
- Configure normal UDP communication between the wired and wireless hosts
- Use the baseline wireless settings for transmission power, bitrate, receiver sensitivity, carrier frequency, path loss, and background noise

2. `Interference_Active`

- Extend the `Interference_Control` configuration
- Activate the `interferer` node by assigning a UDP traffic application
- Configure the `interferer` to generate heavy traffic on the same wireless channel
- Use:
 - large packet sizes
 - short send intervals
 - a delayed start time so interference begins after normal traffic is established
 - a stop time before the simulation ends to observe network recovery

Observe how the presence of a high-traffic interfering node affects throughput, delay, and packet delivery performance in the wireless network.

Observation Table:

Run both configurations. In Qtenv, watch the packet animations and the console log. Fill in the table below for `wirelessHost1`'s traffic (the main victim of interference):

Phase	Time range	Control run	Active interference run
Before interferer starts	$t = 0\text{--}20\text{ s}$		
During interference	$t = 20\text{--}140\text{ s}$		
After interferer stops	$t = 140\text{--}180\text{ s}$		

Table 3: Packet delivery observation table for Task 04

For each cell describe the packet flow as: *normal* / *degraded* / *failed* / *recovering*.

Analysis Questions:

- In Lecture 1 the SINR formula was given as $\text{SINR} = S/(N + I)$. In the `InterferenceActive` run, which term (S , N , or I) changed compared to the control run? Did S (the desired signal) change? Explain.
- After the interferer stopped at $t = 140\text{ s}$, did `wirelessHost1`'s traffic recover? How long did recovery take? What does this tell you about the nature of co-channel interference compared to permanent noise?
- From Lecture 1, the ISM band (2.4 GHz) is used by Wi-Fi, Bluetooth, microwave ovens, and cordless phones. Your `interferer` host models one of these. Which real-world device does it best represent, and why?
- Propose one practical solution (mentioned in Lecture 1) that a network engineer could use to reduce the impact of the interference you simulated. Explain how the solution addresses the underlying problem.

Deliverables

- **Modified NED file:** `Task4_StudentID.ned` (with `interferer` submodule added)
- **INI file:** `omnetpp.ini`
- **Two screenshots:** control run and active interference run, both taken at $t \approx 60\text{ s}$
- **One additional screenshot:** taken at $t \approx 150\text{ s}$ in the active interference run (showing recovery)
- **Completed observation table**
- **Written answers** to (a)–(d)